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A pan-cancer analysis of CTSC as a candidate prognostic and immune-related biomarker

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Abstract

Background Cathepsin C (CTSC) is a lysosomal protease involved in immune regulation and inflammatory responses, with emerging roles in tumor progression and microenvironment remodeling. Although CTSC dysregulation has been observed in several cancers, its pan-cancer significance, immune-related functions, and clinical relevance remain poorly characterized.

Methods We conducted an integrated multi-omics analysis using data from TCGA, GTEx, CPTAC, cBioPortal, HPA, and other public databases. We evaluated CTSC expression patterns, prognostic value, immune infiltration, epigenetic regulation, mutation profiles, and drug sensitivity across diverse cancer types. Analytical methods included survival analysis, functional enrichment, immune correlation assays, and single-cell sequencing.

Results CTSC was widely expressed and significantly dysregulated in multiple cancers. High CTSC expression correlated with poor prognosis in 10 cancer types and was linked to advanced tumor stage. CTSC expression was associated with immune checkpoint genes, infiltration of cancer-associated fibroblasts (CAFs), and $\gamma\delta$ T cells, indicating a dual role in both promoting and suppressing anti-tumor immunity. Additionally, CTSC expression correlated with DNA hypermethylation, RNA methylation regulators, TMB, and MSI. Drug sensitivity analysis suggested that high CTSC expression may enhance response to certain anti-cancer agents.

Conclusion Our study underscores the multifaceted role of CTSC in tumor immunity and progression, supporting its potential as a prognostic biomarker and therapeutic target across cancer types. These findings provide a foundation for further mechanistic and clinical investigation into CTSC-targeted strategies.

Keywords CTSC, Pan-cancer, Prognosis, Immune infiltration, Immunotherapy, Biomarker

1 Introduction

Cancer continues to rank among the foremost causes of death globally [1]. Although considerable progress has been made in early detection and therapeutic interventions, cancer remains a critical public health issue, characterized by persistently high incidence and mortality. In 2020, approximately 19.3 million new cancer cases were reported



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worldwide, alongside close to 10 million fatalities [2]. The development of various cancers involves multigenic and multipathway mechanisms, reflecting a highly complex etiology that incorporates genetic, epigenetic, and environmental factors. These collectively drive tumor initiation, expansion, and metastatic dissemination.

Recent progress in molecular biology and bioinformatics has greatly facilitated the elucidation of cancer mechanisms at the molecular level, leading to the identification of pivotal elements, such as specific receptors and signaling proteins—that exert crucial influences on tumor development, progression, and resistance to treatment [3, 4]. The tumor microenvironment (TME) constitutes a complex ecosystem, with cancer-associated fibroblasts (CAFs) and tumor-infiltrating immune cells as core cellular components. Specifically, CAFs represent a heterogeneous cellular population that plays a pivotal role in TME remodeling; given their substantial influence on tumor progression, gaining enhanced insights into their properties and functional roles is indispensable [5]. Additionally, tumor-infiltrating immune cells significantly impact tumor occurrence and development, either promoting or antagonizing these pathological processes. Furthermore, features of the TME not only serve as reliable markers for assessing tumor cell responses to immunotherapy but also directly influence clinical outcomes. Collectively, systematic investigations into these TME-related components—including CAFs, tumor-infiltrating immune cells, and their associated characteristics—have unveiled novel mechanistic insights and promising therapeutic targets for early cancer diagnosis and personalized treatment strategies [6, 7].

Cathepsin C (CTSC), also known as dipeptidylpeptidase I, is a lysosomal protease expressed in various tissues, particularly inflammatory cells [8]. It centrally regulates immune and inflammatory responses by activating serine proteases in immune cells, including neutrophil serine proteases, granzymes, and tryptases [9]. Mutations in the CTSC gene have been identified as the cause of Papillon-Lefèvre syndrome (PLS) [10]. Traditionally recognized as a lysosomal protease with functions primarily confined to intracellular degradation, CTSC has recently been demonstrated to be frequently over-expressed in various cancers—including non-small cell lung cancer, breast cancer, and colorectal cancer. Studies over the past four years have consistently shown that tumor-secreted CTSC can remotely activate neutrophils, thereby modulating the tumor microenvironment to facilitate tumor growth, metastasis, and immune evasion. This distal regulation further orchestrates the immune cell composition in pre-metastatic sites such as the lungs or within the primary tumor niche [11–13].

However, current investigations into CTSC have largely been confined to a limited range of cancer types, leaving its functional roles in other malignancies insufficiently explored. To comprehensively evaluate the implications of CTSC in oncology, we performed an integrated analysis using multi-source databases—including TCGA, GTEx, cBioPortal, and the Human Protein Atlas (HPA). This study encompasses CTSC expression profiles, association with clinical survival, immune checkpoint correlation, prognostic relevance, immune modulator interactions, genomic alterations, immune microenvironment features, as well as DNA and RNA methylation patterns. Through this pan-cancer approach, we aim to systematically elucidate the potential of CTSC as both a prognostic biomarker and a promising target for therapeutic intervention across diverse cancer types. An overview of the experimental approaches employed herein is provided in Fig. 1.

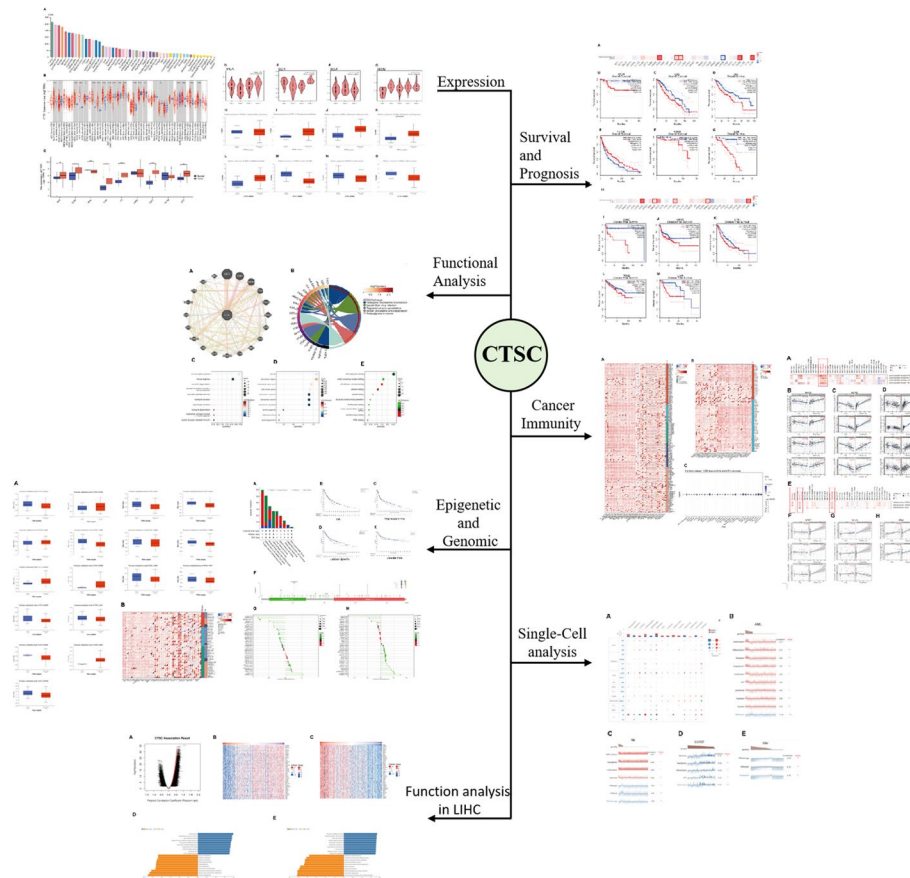


Fig. 1 Schematic overview of the pan-cancer analysis of CTSC

2 Materials and methods

2.1 Analyses of CTSC expression across diverse normal tissues

Transcript abundance data (TPM values) from the Human Protein Atlas (HPA, <https://www.proteinatlas.org/>) and GTEx public databases (<https://gtexportal.org/home/>) were first extracted to quantify CTSC expression levels; these resources were then employed to delineate the distribution and overall expression pattern of CTSC across normal human tissues.

2.2 Gene expression analysis of CTSC in pan-cancer

To assess differential CTSC expression in malignant versus normal tissues, cancer transcriptome datasets were acquired from the UCSC Xena platform (<https://xenabrowser.net/>). All expression values were $\log_2(x + 0.001)$ transformed for normalization prior to subsequent analyses, and the specific abbreviations representing the 33 types of tumors examined in our study are provided in Table S1 for clarity and reference [14]. We further employed GEPIA2 (<http://gepia2.cancer-pku.cn/>) to compare CTSC expression levels across tumor and matched normal samples, and to investigate potential correlations with clinical parameters.

2.3 Cancer expression patterns and pathological staging analysis

To evaluate the expression of CTSC across various cancer types, we integrated data from the TCGA (<https://portal.gdc.cancer.gov/>) and GTEx databases. The GEPIA2 online tool

(<http://gepia2.cancer-pku.cn/>) was utilized to compare CTSC expression levels between tumor and normal tissues, as well as to assess its association with pathological cancer stages. Furthermore, protein expression differences of CTSC in normal versus malignant tissues were examined using samples from TCGA and the CPTAC database.

2.4 Prognostic analysis of CTSC

We utilized GEPIA2 to retrieve survival analyses and generate survival maps for CTSC across all TCGA tumor types, evaluating both Overall Survival (OS) and Disease-Free Survival (DFS). Patients were stratified into high- and low-expression groups based on CTSC levels using predefined thresholds of the upper and lower 50 percentiles. The 50th percentile cutoff was selected to ensure balanced sample sizes across groups and to maintain consistency across tumor types, in line with previous pan-cancer transcriptomic studies [15]. Furthermore, we conducted "optimal cut-off point" survival analysis for 4 types of cancer using the Xiantao Academic Platform, and the results were exactly the same as those of the 50% cut-off point, these validation plots are provided in Supplementary Figure S1.

2.5 Gene functional enrichment analysis

GeneMANIA (<http://genemania.org/>), a publicly accessible platform for studying gene–gene interactions and functional associations, was employed to identify a set of 40 genes co-expressed with CTSC [16]. We subsequently applied the “Similar Gene Detection” module in GEPIA2 to select the top 100 genes correlated with CTSC. Expression levels of the top 10 associated genes across various tumors were acquired via the TIMER2.0 web resource and visualized using heatmaps. Finally, KEGG pathway and GO Biological Process annotations for these targeted and correlated genes were retrieved through the Sangerbox3.0 database (<http://sangerbox.com/home.html>).

2.6 Profiling of immune-regulatory genes and checkpoint molecules

Using data downloaded from the UCSC database, we investigated how CTSC transcript levels correlate with immune-pathway signatures—namely chemokine receptors, MHC molecules, immunoinhibitory and immunostimulatory genes—by processing the files through MuTect2 and R-Project [17]. Pearson’s correlation test was applied to quantify the relationship between CTSC transcript abundance and a 60-gene panel encompassing suppressive and activating immune-checkpoint signatures [18]. These marker genes were extracted from a pan-cancer cohort obtained via the UCSC repository on a per-sample basis.

2.7 Immune infiltration analysis

TIMER2.0 (<http://timer.comp-genomics.org/>) was employed to quantify the association between CTSC transcript levels and immune-cell infiltration within the tumor milieu. Multiple computational tools, such as EPIC, MCPOUNTER, XCELL, and TIDE were applied to estimate the abundance of immune populations across diverse malignancies. Here, we focused on cancer-associated fibroblasts (CAFs) and $\gamma\delta$ T cells, two pivotal subsets implicated in tumor progression and immune evasion. To ensure robustness, we considered a correlation reliable only when ≥ 3 out of 4 algorithms showed consistent and statistically significant results ($p < 0.05$). Discordant results were not reported.

2.8 Pharmacological response profiling

GSCALite (<http://bioinfo.life.hust.edu.cn/web/GSCALite/>; accessed on 8 April 2022) is a web-based integrated platform for the analysis of gene expression profiling and drug sensitivity.

2.9 Analysis of mutation, methylation

To examine the frequency of CTSC mutations and their impact across multiple cancer types, we utilized cBioPortal (<https://www.cbioportal.org/>). Additionally, methylation analyses were conducted using the CPTAC dataset (<https://ualcan.path.uab.edu/analysis-prot.html>) to explore functional effects on CTSC and its role in cancer biology.

2.10 Analysis of CTSC correlation with TMB and MSI indicators

Tumor Mutational Burden (TMB) quantifies the total count of somatic mutations per million bases within tumor tissue. It is well-established that a high TMB can drive the formation of highly immunogenic, tumor-specific neoantigens, and constitutes an emerging predictive biomarker for immunotherapy responsiveness. [19]. Microsatellite instability (MSI) arises from deficiencies in the DNA mismatch repair (MMR) system, resulting in genomic instability and accelerated tumorigenesis, with significant implications for patient prognosis [17, 20, 21]. The associations between CTSC expression and both TMB and MSI were evaluated using Spearman correlation analysis.

2.11 Single-cell sequencing

The Cancer single-cell state Atlas (CancerSEA, <http://biocc.hrbmu.edu.cn/CancerSEA/>) is an analytic tool for studying cancer cell functions at the single-cell level, containing 14 tumor-related cellular functions of 900 cancer cells from 25 cancers [18]. On the basis of single-cell sequencing data, we examined the correlation of CTSC expression and tumor functions. Cell-type-specific CTSC expression was quantified by merging GEO-supplied annotations with the normalized matrix; Wilcoxon test compared malignant vs non-malignant compartments.

2.12 Exploration of CTSC coexpression network

The LinkedOmics platform (<http://www.linkedomics.org/login.php>) provides a publicly available web-based interface for the analysis of multi-dimensional gene expression data [22, 23]. In this investigation, we employed LinkedOmics to detect genes exhibiting co-expression with CTSC using Pearson's correlation analysis. The outcomes were graphically represented through volcano plots and heatmaps. Additionally, Gene Ontology biological process (GO-BP) and pathway enrichment analyses pertaining to CTSC were conducted to elucidate its functional implications.

3 Results

3.1 Expression levels analysis of CTSC in pan-cancer

Evaluation of the HPA and GTEx databases revealed that CTSC is broadly expressed throughout normal human tissues, exhibiting particularly high abundance in adipose tissue, lung, placenta, and lymph nodes. (Fig. 2A). Evaluation of CTSC mRNA expression within the TCGA database revealed significant upregulation in tumors including HNSC and LIHC [24], while marked downregulation was observed in LUAD and

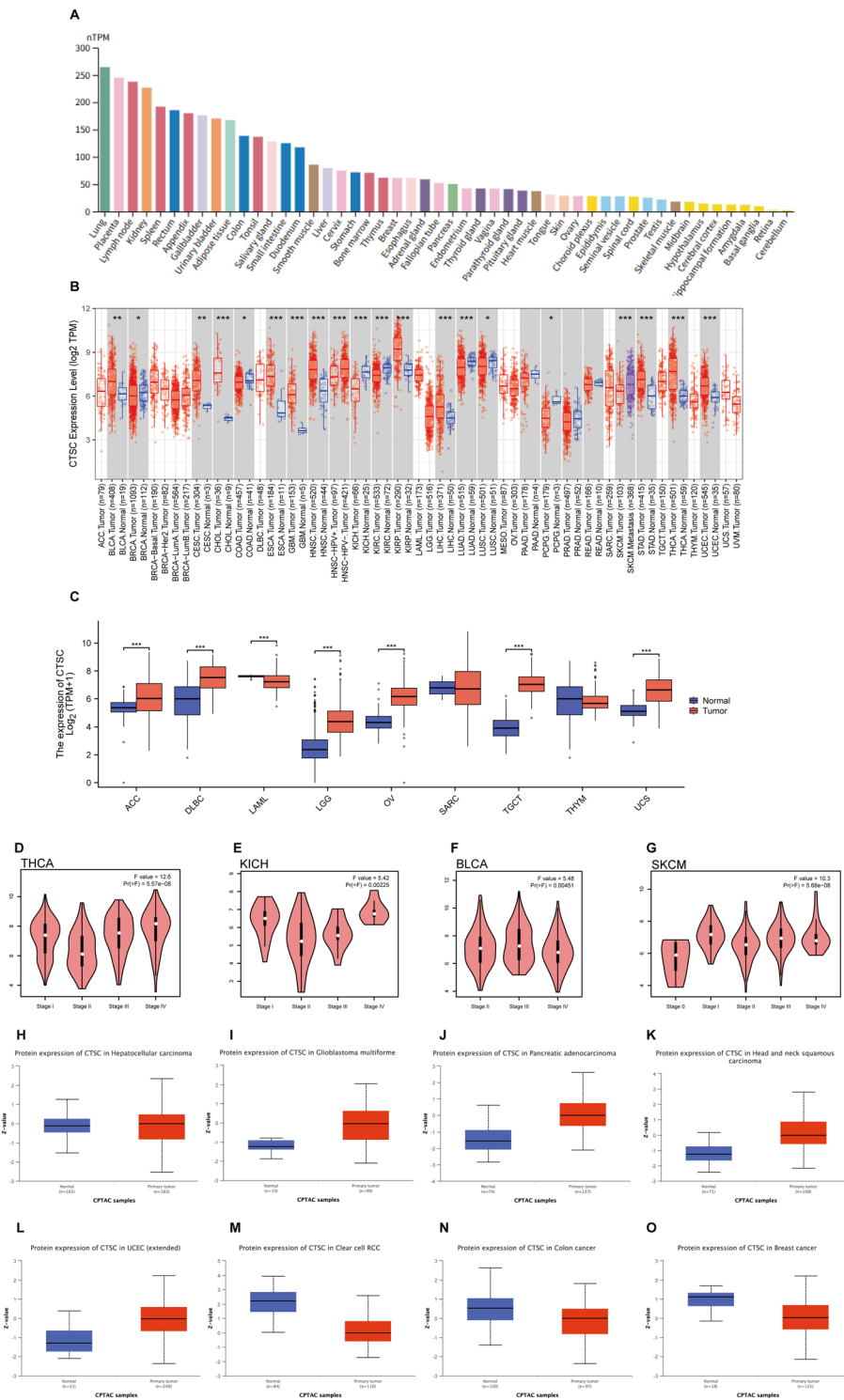


Fig. 2 Dysregulation of CTSC and its association with Pathological Stages Across Cancers. **A** CTSC is broadly expressed in normal human tissues (HPA database). **B** The expression levels of CTSC across various cancer types (TCGA database). **C** CTSC expression across in ACC, DLBC, LAML, LGG, OV, TGCT, UCS (TCGA and GTEx database). **D–G** Changes in CTSC mRNA expression across cancer stages (I to IV) were analyzed for THCA, KICH, BLCA, and SKCM using TCGA data. **H–O** The pattern of CTSC protein expression was assessed in selected cancers using data derived from CPTAC samples. (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

LUSC (Fig. 2B). Further integrated analysis of TCGA and GTEx datasets identified elevated CTSC expression in additional malignancies such as ACC, DLBC, LAML, LGG, OV, TGCT, and UCS (Fig. 2C). Pathological stage-based examination indicated substantial variations in CTSC expression at different stages of cancers like BLCA, KICH, THCA, and SKCM, though no significant differences were detected in other cancer types (Fig. 2D–G). Additionally, protein-level data from the CPTAC database confirmed increased CTSC expression in GBM, pancreatic adenocarcinoma, head and neck squamous cell carcinoma, and UCEC, but decreased levels in COAD, BRCA, and clear cell renal cell carcinoma (Fig. 2H–O).

3.2 Prognostic role of CTSC expression across cancer types

This study investigated the variation of CTSC expression among diverse cancer types and its association with patient prognosis by analyzing TCGA data through the GEPIA2 platform. In cancers such as KICH, LGG, LIHC, THYM, and UVM, elevated CTSC expression was correlated with reduced overall survival, indicating a potential role of CTSC in worsening clinical outcomes in these malignancies (Fig. 3A–G). Conversely, in SKCM, low CTSC expression was linked to shorter survival, suggesting a possible protective function or enhanced treatment responsiveness associated with CTSC in this context. Furthermore, high CTSC levels were related to shorter disease-free survival (DFS) in several cancers—including DLBC, HNSC, LGG, UVM, and PRAD—reinforcing the notion that CTSC may contribute to unfavorable prognosis (Fig. 3H–M).

3.3 Functional enrichment analysis of CTSC

To explore the potential involvement of CTSC in tumor development, we retrieved 20 genes linked to CTSC from the GeneMANIA database (Fig. 4A). Furthermore, using GEPIA2, we extracted 100 genes correlated with CTSC from the TCGA database. Subsequent enrichment analysis demonstrated that CTSC is associated with five key signaling pathways: pathogenic *Escherichia coli* infection, Epstein-Barr virus infection, regulation of actin cytoskeleton, antigen processing and presentation, and proteoglycans in cancer (Fig. 4B). Gene Ontology (GO) analysis suggested a role for CTSC in immune system processes and immune response (Fig. 4C). Cellular component analysis revealed that CTSC-related genes are primarily enriched in vesicles, extracellular regions, extracellular space, exosomes, and other extracellular organelles (Fig. 4D). Molecular function analysis indicated involvement in identical protein binding (Fig. 4E).

3.4 CTSC expression links immune modulators, checkpoint molecules, and chemosensitivity profiles

Subsequent analyses focused on the associations of CTSC expression with immune-modulating genes, immune checkpoint molecules, and drug sensitivity. Using a pan-cancer dataset obtained from the UCSC database, we extracted expression data for CTSC along with 150 marker genes, applying stringent filtering criteria to ensure data quality for each sample. Pearson correlation analysis was performed to evaluate the relationship between CTSC and four categories of immune pathway markers. Intriguingly, CTSC expression showed a positive correlation with most immunomodulatory genes (Fig. 5A). Further assessment revealed significant correlations between CTSC and multiple immune checkpoint molecules (Fig. 5B). Moreover, drug sensitivity analysis

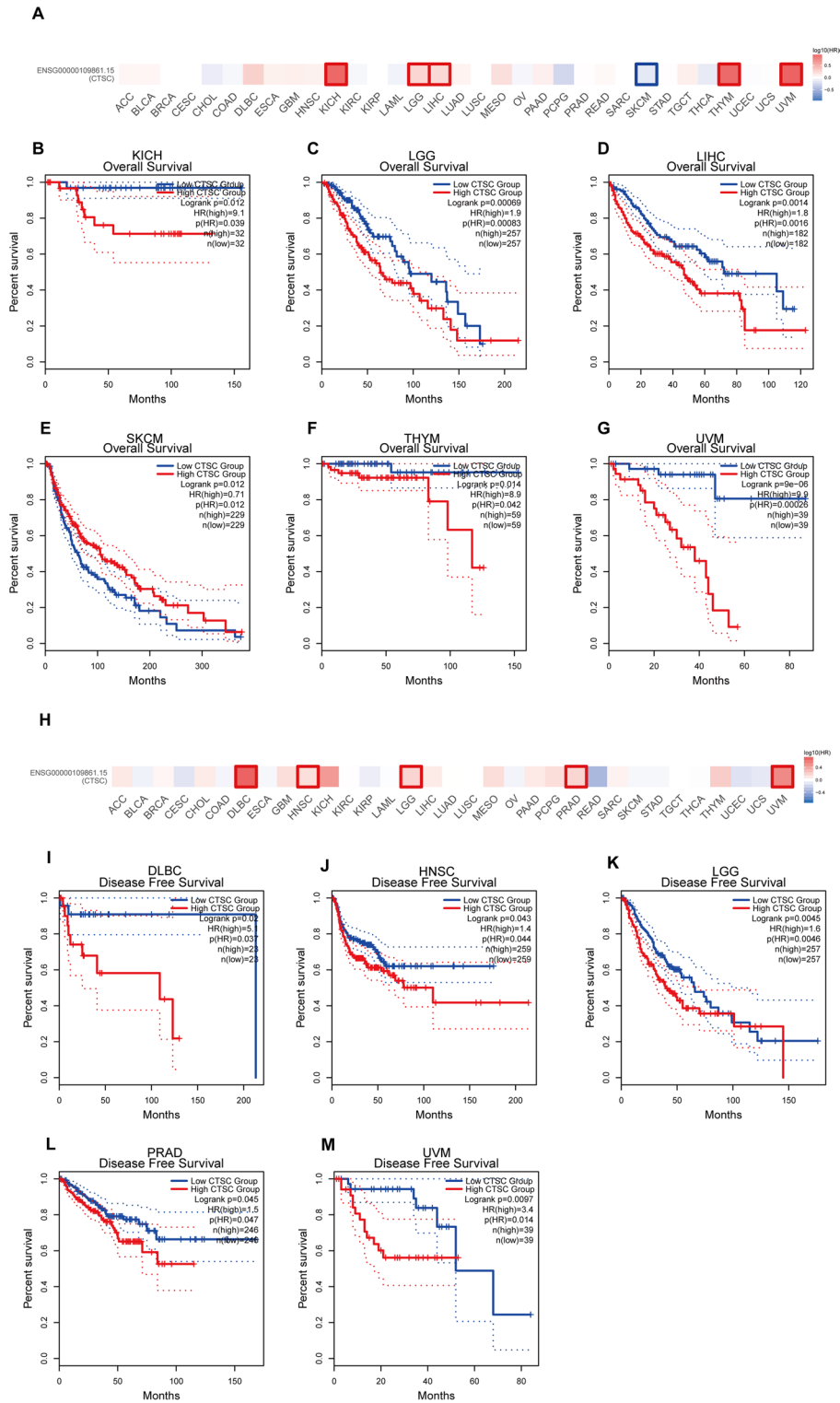


Fig. 3 Survival Correlation of CTSC Expression Across Cancers. **A–G** Association between CTSC expression levels and overall survival (OS). **H–M** Disease-free survival (DFS) in multiple cancer types. Analyses were performed with the GEPIA2 platform using TCGA data

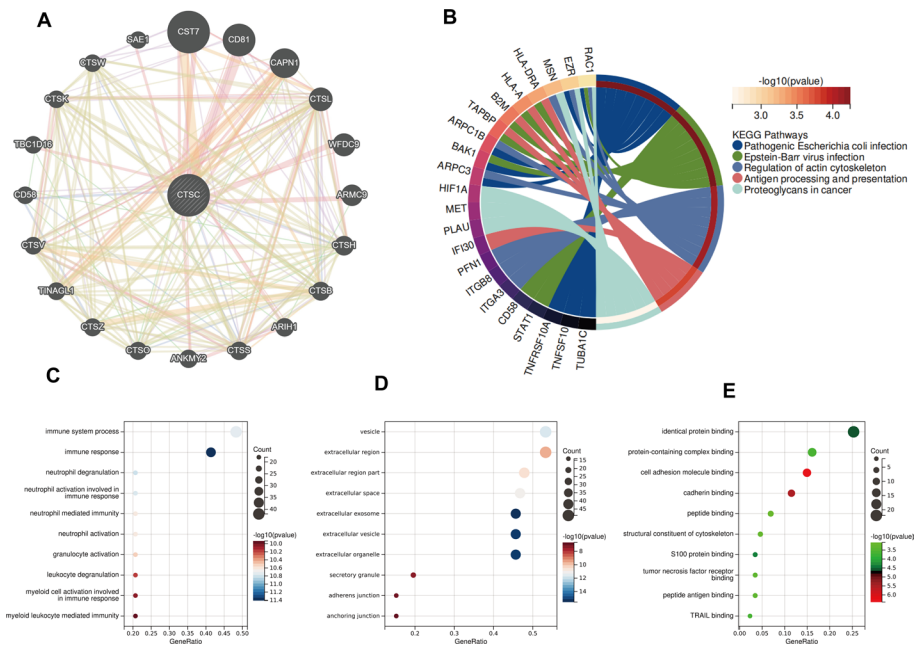


Fig. 4 Analysis of CTSC-Related Gene Enrichment. **A** network of twenty genes co-expressed with CTSC from GeneMANIA. **B** Pathway enrichment of CTSC-linked genes. **C–E** Gene Ontology enrichment results for biological processes (**C**), cellular components (**D**), and molecular functions (**E**)

conducted through GSCALite demonstrated a consistent inverse relationship between CTSC expression levels and the IC_{50} values of the tested antitumor agents (Fig. 5C), Importantly, 10 of the 30 drugs showing CTSC-associated sensitivity are already FDA-approved and recommended by NCCN for the exact cancer types where high CTSC predicted lower IC_{50} (Supplementary Table S2), underscoring the immediate clinical relevance of the signature. The remaining agents are clearly identified as experimental, guiding future prioritisation.

3.5 The impact of CTSC on tumor immune infiltration: modulation of fibroblasts and $\gamma\delta$ T cells

In this investigation, we employed a range of bioinformatic algorithms to assess the association of CTSC expression with two critical elements of the tumor microenvironment cancer-associated fibroblasts (CAFs) and $\gamma\delta$ T cells—to uncover potential pathways through which CTSC could affect cancer development. Leveraging the TIMER2.0 resource integrated with computational methods such as EPIC, MCP-COUNTER, XCELL, and TIDE, we analyzed correlations between CTSC transcript abundance and CAF infiltration levels using TCGA datasets. Our analysis revealed a marked positive association between high CTSC expression and increased CAF prevalence in several malignancies (Fig. 6A). Notably, strong relationships were identified in PAAD, PCPG, and PRAD (Fig. 6B–D), indicating that CTSC might drive oncogenesis by influencing CAF behavior, remodeling the tumor microenvironment, and potentially promoting immune escape. We further explored the link between CTSC expression and $\gamma\delta$ T cell infiltration (Fig. 6E). By implementing analytical approaches including CIBERSORT, CIBERSORT-ABS, and XCELL, a statistically significant positive correlation was detected in MESO, UVM, and THYM, wherein elevated CTSC levels were associated with enhanced $\gamma\delta$ T cell presence (Fig. 6F–H). These observations suggest that CTSC

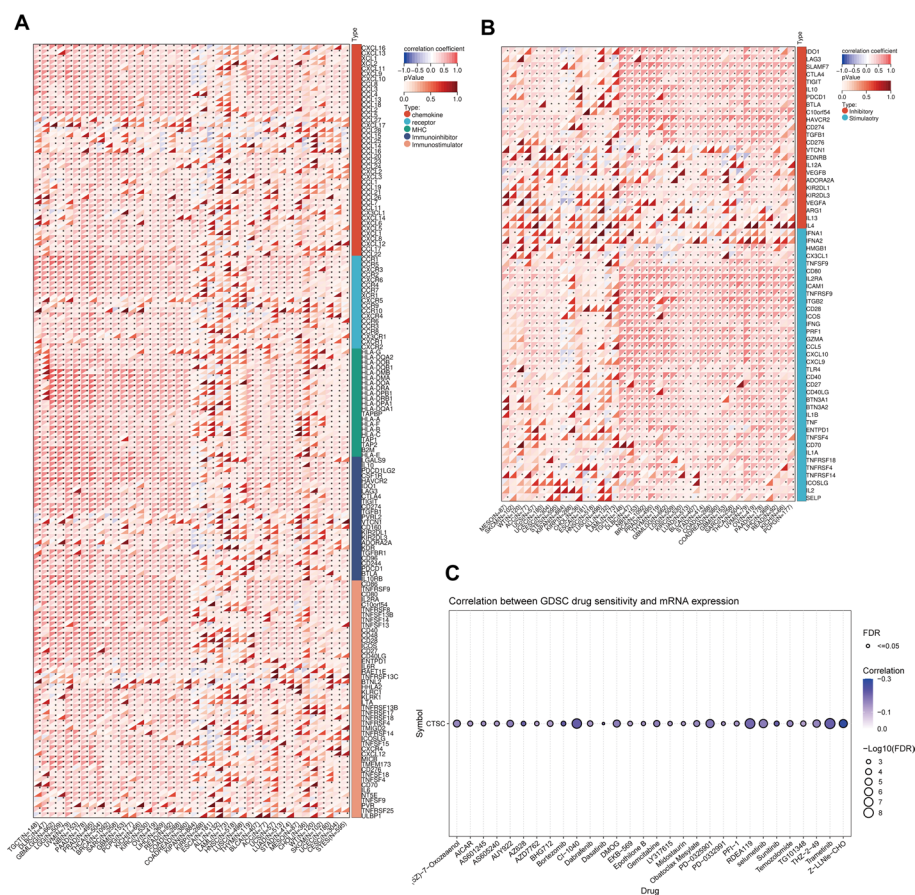


Fig. 5 Interrogation of CTSC Correlations with Immunoregulatory Molecules and Therapeutic Sensitivity. **A** Systematic profiling of CTSC and immune regulatory gene co-expression. **B** Pan-cancer analysis of CTSC linkage with established immune checkpoint genes in TCGA. **C** Assessment of CTSC-dependent chemosensitivity using the GSCALite platform. (* $p < 0.05$)

could potentially exert anti-tumor functions in these specific cancers through the activation or recruitment of $\gamma\delta$ T cells.

3.6 The influence of CTSC gene mutations on tumor biology and patient prognosis across various cancer types

Analysis of over 500 cancer samples using cBioPortal revealed significant mutation frequencies in cancers such as head and neck squamous cell carcinoma, uterine corpus endometrial carcinoma, and ovarian serous cystadenocarcinoma. Among the 32 cancer types examined, ovarian serous cystadenocarcinoma exhibited the highest frequency of CTSC alterations, reaching 6% (Fig. 7A). For context, the 6% alteration frequency of CTSC in OV is comparable to the median frequency (5.8%) of the 158 significantly mutated genes reported in the same TCGA-OV cohort [25]. Most pathogenic mutations of CTSC are located within its catalytic domain (Peptidase_C1) or its propeptide/light chain region (corresponding to the exon segments covered by CathepsinC_exc). Furthermore, the F105L missense mutation in CTSC, although uncommon in the general cancer population, has been implicated in Papillon-Lefèvre syndrome, suggesting a potential biological role [26]. R182Q reduces activity to 50%, while C285R disrupts the Cys285-Cys325 disulfide bond and retains <10% activity. Thus, the CTSC mutations detected

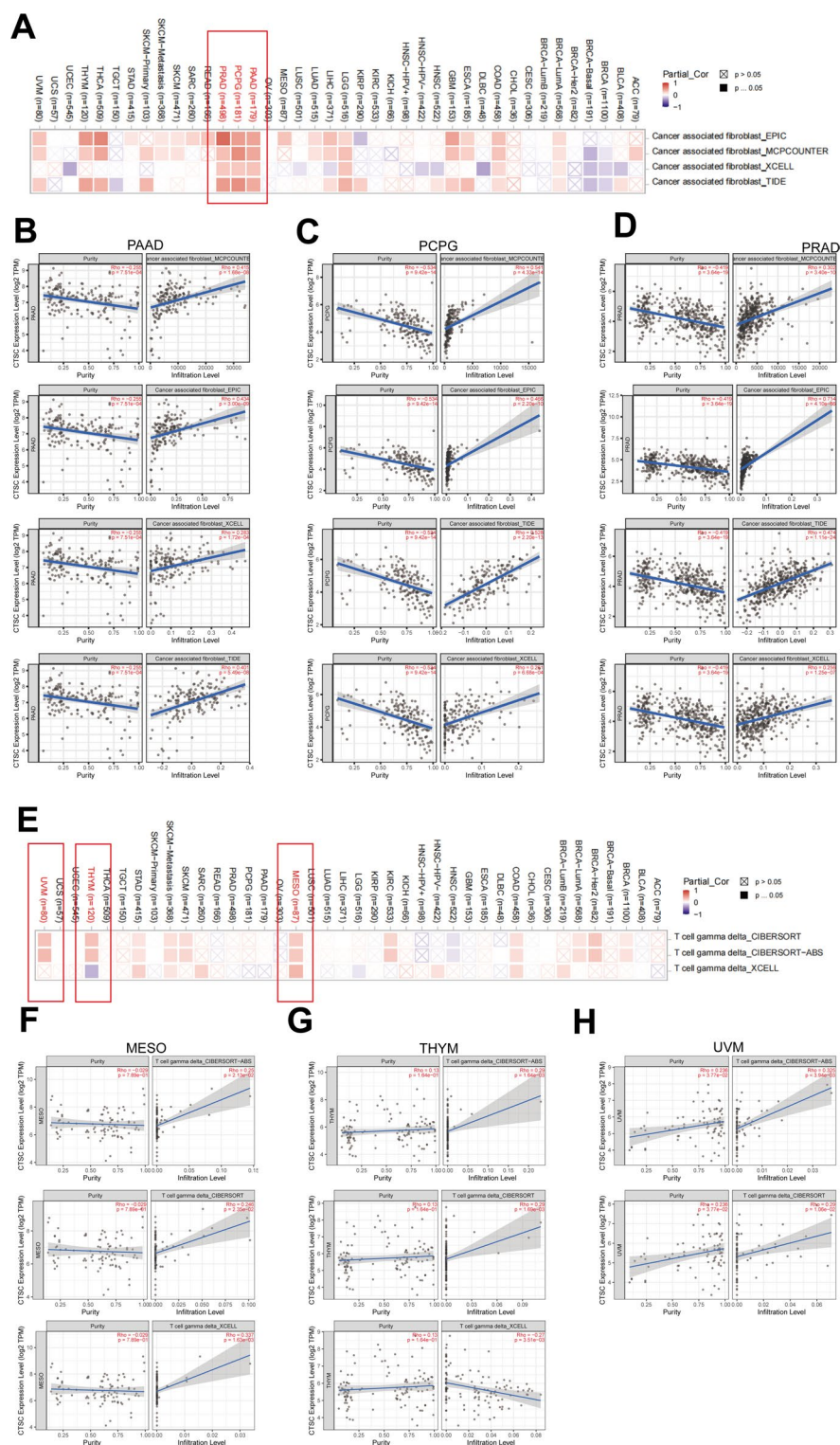


Fig. 6 Assessment of CTSC Expression and Immune Cell Infiltration in the Tumor Microenvironment. **A** The association between CTSC transcript levels and cancer-associated fibroblast (CAF) infiltration was evaluated using the EPIC, MCP-COUNTER, XCELL, and TIDE algorithms in TIMER 2.0, based on TCGA data. **B–D** Correlation scatter plots of CTSC expression and CAF infiltration in PAAD (B), PCPG (C), and PRAD (D), as quantified by the aforementioned algorithms. **E** Infiltration levels of $\gamma\delta$ T cells were analyzed against CTSC expression using CIBERSORT, CIBERSORT-ABS, and XCELL methods. **F–H** Scatter plots demonstrating the relationship between CTSC expression and $\gamma\delta$ T cell infiltration in MESO (F), THYM (G), and UVM (H)

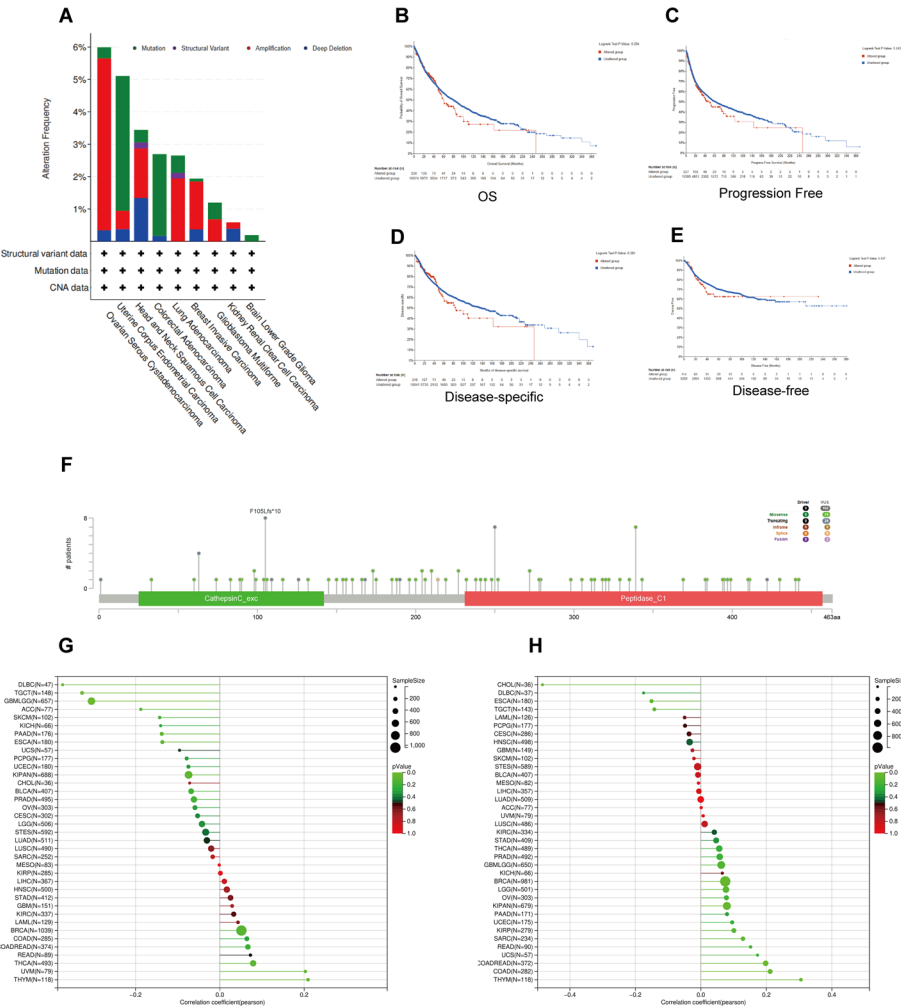


Fig. 7 We characterized CTSC gene mutations across various malignancies. **A** Using cBioPortal and TCGA, we quantified the frequency of CTSC alterations, encompassing mutations, structural variants, amplifications, and deletions. **B–E** The impact of CTSC mutation status on patient survival outcomes (OS, PFS, DSS, DFS) was assessed using Kaplan–Meier plots and Logrank tests. **F** A recurrent mutation site in CTSC and its incidence. **G–H** The association of CTSC genetic status with TMB (**G**) and MSI (**H**) was also examined

across our pan-cancer cohort are functionally deleterious and are expected to impair downstream protease activation and immune-modulatory signaling [27]. The occurrence of CTSC mutations across these malignancies may reflect their involvement in tumor development. Our findings detail the spectrum, genomic distribution, and prevalence of CTSC mutations, with missense mutations representing the most common alteration type (Fig. 7F). Survival analyses-including overall survival (OS), disease-free survival (DFS), disease-specific survival (DSS), and progression-free survival (PFS)-revealed no statistically significant differences between groups with and without CTSC mutations (Fig. 7B–E; log-rank $P > 0.05$). These results imply that the effect of CTSC mutations on clinical outcomes may be context-dependent, influenced by cancer type or coexisting molecular alterations. We also assessed the relationship between CTSC expression and tumor mutational burden (TMB), both considered predictive biomarkers for immunotherapy response. Among 37 tumor types analyzed, eight cancers including GBMLGG, KIPAN, BLCA, PRAD, LGG, STEC, LUAD, and LUSC, which exhibited a significant inverse correlation between TMB and CTSC expression (Fig. 7G). Additionally, CTSC

expression was associated with microsatellite instability (MSI) in several cancers: positive correlations were detected in KIPAN, BRCA, and GBMLGG, whereas negative correlations were observed in HNSC, STEC, BLCA, and LIHC (Fig. 7H).

3.7 Association of CTSC expression with DNA and RNA methylation modifications

This study examined the correlation between CTSC expression and epigenetic modifications involving both DNA and RNA methylation. Based on data obtained from the UALCAN database, DNA methylation levels of CTSC were evaluated across multiple cancer types. The analysis revealed markedly elevated methylation of CTSC in most malignancies relative to normal tissues, potentially contributing to its upregulated expression in tumors (Fig. 8A). Furthermore, positive associations were observed between CTSC and critical regulators of m1A, m5C, and m6A RNA modifications in several cancers, implying that altered expression of RNA methylation-related genes may participate in the oncogenic functions mediated by CTSC (Fig. 8B).

3.8 Single-cell functional analysis of CTSC

To delineate the potential multifaceted role of CTSC in tumorigenesis, we performed a single-cell transcriptional profiling analysis via the CancerSEA platform (Fig. 9A). In

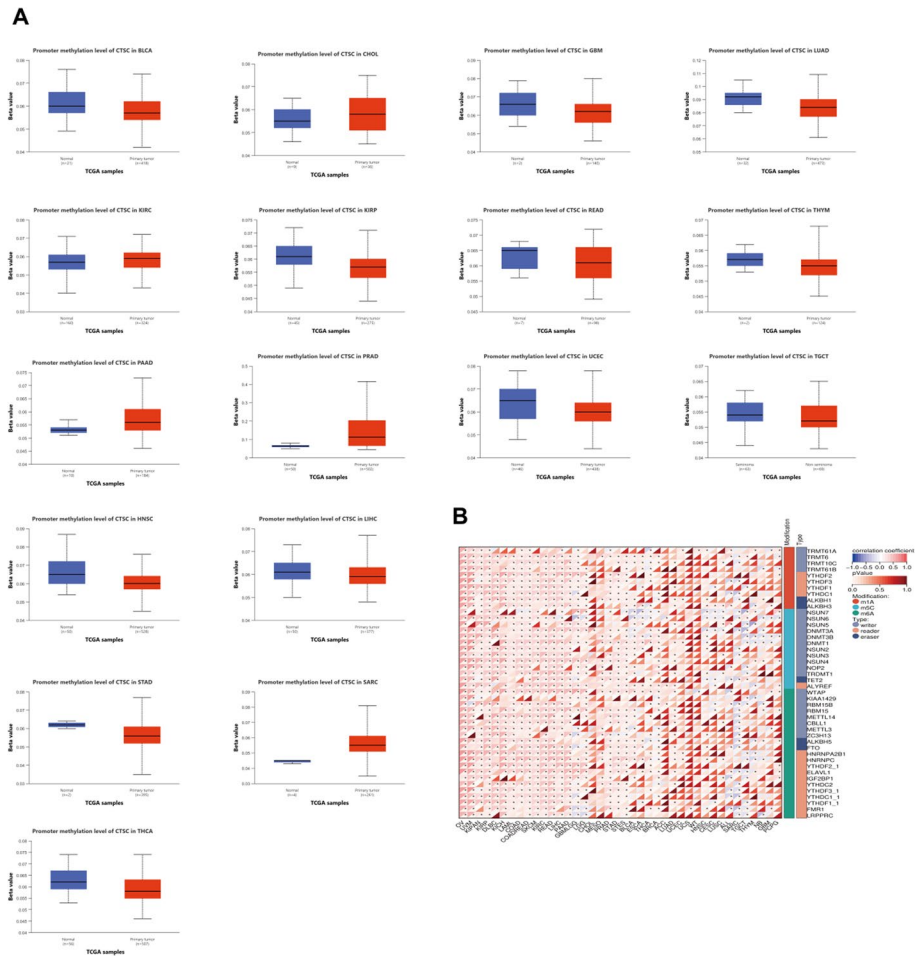


Fig. 8 CTSC facilitated oncogenesis by elevating gene methylation levels. **A** Assessment of CTSC DNA methylation conducted using UALCAN. **B** Correlation of CTSC expression with genes involved in RNA methylation

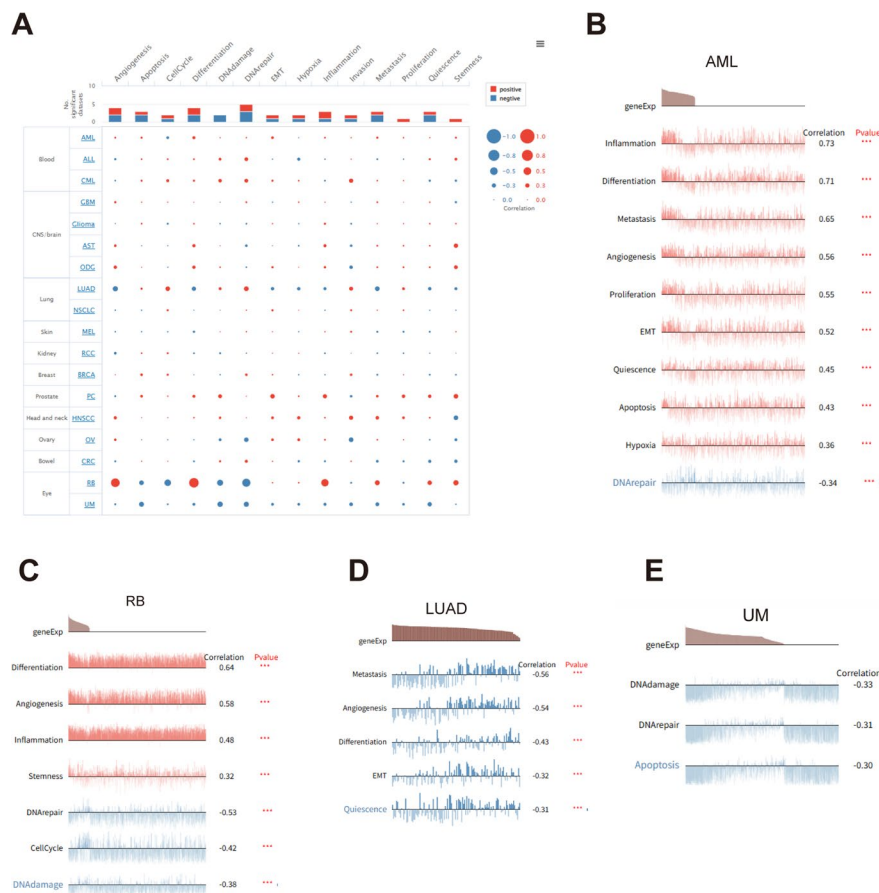


Fig. 9 CTSC's functional annotation role for single-cell data in the CancerSEA repository. **A** Functional status of CTSC in different human cancers. **B–E** Correlation analysis between functional status and CTSC in UM, LUAD, AML, and RB. (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

AML, CTSC positively correlated with inflammation and angiogenesis, indicating a pro-tumorigenic role in leukemic contexts. In RB, CTSC showed negative correlations with DNA damage and DNA repair, but positive correlations with differentiation and angiogenesis, hinting at a dual role in tumor maturation and vascularization. In LUAD, CTSC negatively correlated with differentiation and angiogenesis, suggesting a role in maintaining undifferentiated and anti-angiogenic states. In UM, CTSC negatively correlated with DNA damage and DNA repair, suggesting a role in genomic instability (Fig. 9B–E). To determine the cellular source of CTSC within the tumour microenvironment, we re-analysed the same four scRNA-seq datasets. We have now clarified the cellular source of CTSC by re-analysing the four CancerSEA scRNA-seq datasets (AML, LUAD, RB, UM) after downloading their raw matrices and cell-type labels from GEO. CTSC was markedly enriched in malignant cells (mean TPM 4.7–6.2) and > fivefold lower in immune or stromal subsets ($p < 0.001$, Wilcoxon; Supplementary Table S3), indicating that the previously mapped functional phenotypes are tumour-cell-intrinsic.

3.9 Investigating CTSC function in LIHC patients

Building upon these results, CTSC demonstrates a substantial connection with both cancer immunity and patient prognosis. To further elucidate its functional relevance in tumor contexts, we employed the LinkedOmics database to characterize the

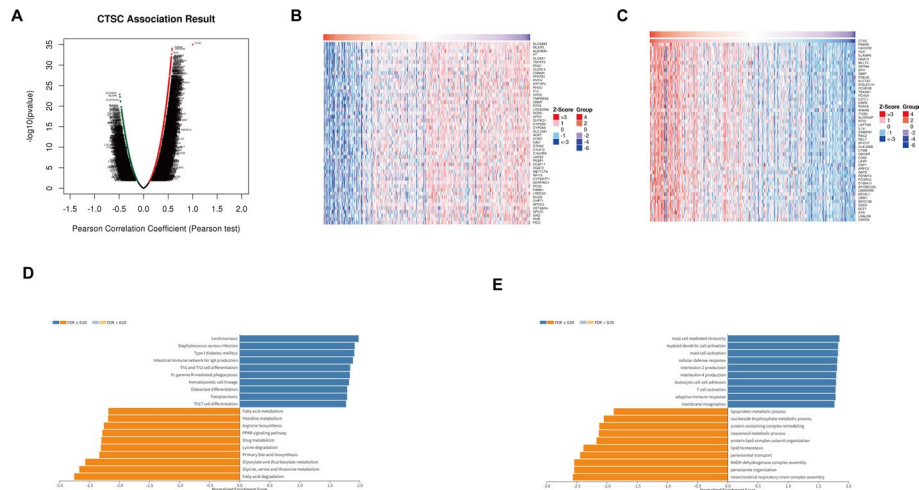


Fig. 10 Functional and pathway enrichment profiling for CTSC-coexpressed genes in LIHC. **A** Correlation analysis of CTSC-coexpressed genes based on Pearson test in the LIHC cohort. **B** Heatmap of the top 50 positively correlated genes with CTSC. **C** Heatmap of the top 50 negatively correlated genes with CTSC. **D** KEGG pathway analysis of CTSC-coexpressed genes. **E** Gene Ontology biological process (GO-BP) analysis of CTSC-coexpressed genes

co-expression network of CTSC. In LIHC, genes strongly positively correlated with CTSC are represented by dark red dots, while those exhibiting strong negative correlation are depicted in dark green (Fig. 10A). Heatmaps illustrate the top 50 genes most positively and negatively associated with CTSC, respectively (Fig. 10B, C). KEGG pathway enrichment analysis highlighted significant involvement in multiple immune and infectious processes, including Leishmaniasis, Staphylococcus aureus infection, Type I diabetes mellitus, intestinal immune network for IgA production, Th1 and Th2 cell differentiation, Fc gamma R-mediated phagocytosis, hematopoietic cell lineage, osteoclast differentiation, toxoplasmosis, and Th17 cell differentiation (Fig. 10D). Furthermore, Gene Ontology biological process (GO-BP) terms indicated that CTSC-coexpressed genes are primarily associated with immune-related processes such as mast cell-mediated immunity, activation of myeloid dendritic cells, mast cell activation, cellular defense response, production of interleukins-2 and interleukins-4, leukocyte cell–cell adhesion, T cell activation, adaptive immune response, and membrane invagination (Fig. 10E). Collectively, these findings imply that the co-expression network centered on CTSC may play an important role in immune activation and prognostic outcomes in LIHC.

4 Discussion

The concept that cancer therapies should be tailored to individual patients stems from observed differences in tumor biology and treatment efficacy, which are modulated by hereditary, contextual, and behavioral traits [28]. A multifactorial assessment of the patient's unique circumstance is paramount for devising an optimal treatment regimen. This evaluation must consider tumor characteristics—including its type, grade, and dissemination—as well as the patient's general health and comorbidities [29, 30]. Progress in scientific and technological research has facilitated a deeper comprehension of cancer biology. This has opened up new treatment options. Three areas of interest are ICB treatments, Cytokine treatments and pan-cancer analysis [23, 24], with CTSC playing an important role that is also involved in cancer development and treatment.

We initially assessed CTSC expression across 33 cancer types and matched normal tissues. CTSC is broadly expressed in normal human tissues; however, the physiological implications of its enrichment remain to be elucidated. The results indicated that CTSC was significantly overexpressed in 27 of these cancers, which is consistent with previous finding [31,32]. Furthermore, in several tumor types, CTSC expression levels showed a notable correlation with advanced clinical stages. Analysis via UALCAN revealed elevated CTSC protein expression in UCEC, COAD, and BRCA. These results indicate that CTSC facilitates tumor initiation and progression in various human malignancies. Collectively, these data imply that CTSC acts as a promoter of tumorigenesis and disease advancement across multiple cancer types. These findings support an oncogenic role for CTSC in diverse human cancers. Additionally, high CTSC expression was associated with poorer prognosis in 10 cancer types, indicating its potential utility as a prognostic biomarker. Finally, we found that CTSC and its co-expressed genes are involved in cancer-related and immune-related pathways, including immune system processes and immune response.

Immune checkpoint genes play a crucial role as targets for immune checkpoint inhibitors (ICIs) in treating a variety of cancers [33]. Currently, ICIs have proven to be an effective immunotherapy for cancer [34]. Furthermore, CTSC expression is strongly positively correlated with several immune checkpoint molecules typically found in tumors, indicating that CTSC could be a potential new target for cancer immunotherapy. Our study further revealed substantial crosstalk between CTSC and multiple immune-related genes, implying a potential connection between CTSC expression and the extent of immune cell infiltration within tumors. These interactions could critically influence clinical prognosis and open new avenues for designing immunomodulatory treatments. Furthermore, leveraging data obtained from the GDSC project [35], we analyzed the correlation between CTSC expression and responsiveness to various chemotherapeutic agents. The results demonstrated that elevated CTSC expression correlates with reduced IC50 measures for a panel of 30 distinct anti-cancer compounds. These drugs may exhibit enhanced efficacy in cancers with high CTSC expression. These results provide valuable insights for selecting appropriate drugs in clinical settings and predicting patient responses to treatment. To move from association to prediction, the following pipeline is recommended: (i) Engineer stable CTSC-overexpression and CRISPR-knock-out lines in LUAD, LIHC and BRCA; (ii) Perform 96-well dose-response curves for the 10 FDA-approved compounds highlighted in our analysis; (iii) Calculate Δ IC50 and synergy indices; (iv) Validate findings in PDX or patient-derived organoid models to confirm clinical translatability.

Our study reveals a context-dependent dual immunomodulatory role for CTSC within the tumor microenvironment. On the one hand, CTSC expression positively correlated with immunosuppressive components such as cancer-associated fibroblast (CAF) infiltration and M2 macrophage polarization (Fig. 5A–D), consistent with prior reports that CTSC promotes neutrophil extracellular trap (NET) formation and facilitates immune evasion in cancers such as breast and colorectal carcinoma [9, 10]. We speculate that in immunologically “cold” tumors, CTSC may enhance protease-activated receptor (PAR)-mediated signaling or activate pro-tumorigenic cytokines (e.g., IL-1 β), thereby fostering an immunosuppressive niche and facilitating tumor progression. On the other hand, CTSC exhibited a strong positive correlation with $\gamma\delta$ T cell infiltration in specific

cancer types, including MESO, UVM, and THYM (Fig. 5F–H). Given that $\gamma\delta$ T cells are known to exert potent anti-tumor cytotoxicity under appropriate cytokine conditions, we hypothesize that CTSC may process and activate chemokines (e.g., CCL2, CCL3, CCL4) or surface ligands that promote the recruitment or activation of $\gamma\delta$ T cells, suggesting a potential immune-stimulatory function in selected contexts. This functional duality may reflect tissue-specific microenvironmental factors or differential expression of CTSC substrates or inhibitors, underscoring its complex role in shaping immune responses across different malignancies. We hypothesize that the dual immunomodulatory roles of CTSC may stem from cancer-type-specific microenvironmental contexts. In immunologically 'cold' tumors, CTSC may facilitate CAF-mediated immune suppression, whereas in tumors with active $\gamma\delta$ T cell infiltration, CTSC may enhance anti-tumor immunity via chemokine processing. This functional divergence may also depend on the cellular source of CTSC and its substrate availability.

CTSC mutations occur at frequencies similar to those of established but low-prevalence drivers such as ARID1A in OV (6–8%). However, unlike ARID1A, CTSC mutations showed no significant impact on survival, suggesting that either (1) they represent passenger events, or (2) their effect requires co-occurring alterations (e.g., TP53, PIK3CA). Owing to the low event number, we could not perform reliable co-mutation analysis; this warrants expanded sequencing cohorts.

DNA methylation represents a fundamental epigenetic mechanism that serves a key function in modulating gene expression [36]. However, abnormal DNA methylation patterns are commonly found in cancer cells and can lead to the activation of oncogenes or silencing of tumor suppressor genes, thereby disrupting normal biological functions and promoting uncontrolled cellular proliferation [37]. Moreover, epigenetic analyses revealed that CTSC expression is associated with DNA hypermethylation and RNA methylation regulators, indicating a layer of epigenetic control that may amplify its oncogenic effects [38].

Tumor Mutational Burden (TMB) quantifies the total number of mutations within a tumor genome. Elevated TMB levels are often linked to improved immune recognition of tumor cells, thereby increasing their susceptibility to immunotherapeutic interventions [39], conversely, arises from defects in DNA mismatch repair systems, leading to a hypermutated state. MSI-high tumors typically exhibit heightened mutation loads, which can promote the generation of tumor-specific neoantigens and enhance immunogenicity [40]. The clinical approval of immune checkpoint inhibitors, including nivolumab and pembrolizumab, for MSI-high or mismatch repair-deficient (dMMR) solid tumors has marked a transformative advancement in oncology, offering effective targeted treatment for patients harboring these molecular characteristics [41]. Our study further suggests that CTSC may play a role in modulating the immune microenvironment. This insight opens avenues for developing new targeted therapies against specific cancer types, which could have considerable clinical implications. Alterations in CTSC expression were associated with changes in the immune landscape, highlighting its potential as a target for cancer immunotherapy.

At the single-cell level, CTSC expression was associated with functional states including suppression of DNA damage response and modulation of differentiation pathways, underscoring its contribution to the maintenance of tumorigenic phenotypes. Collectively, these findings highlight the context-dependent functional impact of CTSC, which

variably influences DNA integrity, inflammatory signaling, and developmental processes in a cancer-type-specific manner. Finally, in LIHC, CTSC-centered co-expression networks were enriched in immune and inflammatory pathways, reinforcing its relevance in cancer-related immunity and suggesting its potential as a therapeutic target for HCC [42].

Despite the breadth of our analysis, several limitations must be acknowledged. First, although we utilized multiple public databases to ensure robustness, all findings are derived from bioinformatic analyses and require further validation through *in vitro* and *in vivo* experiments. The lack of functional assays means that causal relationships cannot be firmly established—associations described here remain speculative without mechanistic confirmation. Second, clinical data heterogeneity across different cancer types and sources may introduce biases in survival and correlation analyses. Variations in sample size, sequencing platforms, and patient demographics could affect the generalizability of the results. Third, while immune infiltration algorithms such as TIMER2.0 and CIBERSORT provide valuable insights, they are computational inferences based on transcriptomic data and may not fully represent the actual cellular composition of the tumor microenvironment. Lastly, the study primarily focuses on bulk and single-cell transcriptomics without integrating proteomic validation from large-scale cohorts. Since mRNA levels do not always correspond with protein activity, future studies should include immunohistochemistry or mass spectrometry-based protein expression profiling to reinforce our conclusions.

To address these limitations and validate the bioinformatic insights and mechanistic hypotheses proposed, we recommend the following experimental approaches: *In vitro* studies could include CTSC knockdown and overexpression in relevant cancer cell lines (e.g., derived from LUAD, BRCA, or LIHC) to evaluate its influence on proliferation, invasion, and cytokine secretion. Co-culture systems incorporating fibroblasts or immune cells, such as neutrophils or $\gamma\delta$ T cells could further elucidate CTSC-mediated intercellular communication. *In vivo* validation could employ CTSC-knockout models or xenografts treated with CTSC inhibitors (e.g., AZD7986) to examine how CTSC perturbation affects tumor growth, metastatic potential, and immune infiltration. Subsequent single-cell RNA sequencing of treated tumors could provide detailed characterization of altered immune cell landscapes and functional states. Furthermore, proteomic and mechanistic assays, such as immunoprecipitation followed by mass spectrometry could help identify CTSC-interacting proteins and substrates across tumor and stromal compartments. Additional experiments might assess CTSC-dependent processing and activation of immune modulators (e.g., chemokines and cytokines) via techniques such as Western blot or ELISA. Through these integrated efforts, the predicted roles of CTSC in tumor immunity and progression can be more definitively established.

5 Conclusions

Our multi-faceted bioinformatic investigation establishes CTSC as a key player in cancer immunity and progression across diverse tumor types. It highlights the potential of CTSC as a prognostic marker and therapeutic target, while also laying the groundwork for subsequent functional and clinical studies aimed at clarifying its mechanistic roles and translational value.

Supplementary Information

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Additional file 1.

Additional file 2.

Additional file 3.

Additional file 4.

Author contributions

The authors declare their contributions to the paper as follows: conceptualization and design of the study: Lan Zheng, Xin Liu; data acquisition: Yiling Xi, Bin Ge; analysis and interpretation of findings: Xing Wei, Jinwen Cai, Daicai Gong; preparation of the initial draft: Peng Chen. All authors examined the results and endorsed the final version of the manuscript.

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Data availability

All the necessary data for assessing the conclusions of the paper can be found within the paper itself and its Supplementary Materials.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Author approved the manuscript for publication.

Competing interests

The authors declare no competing interests.

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